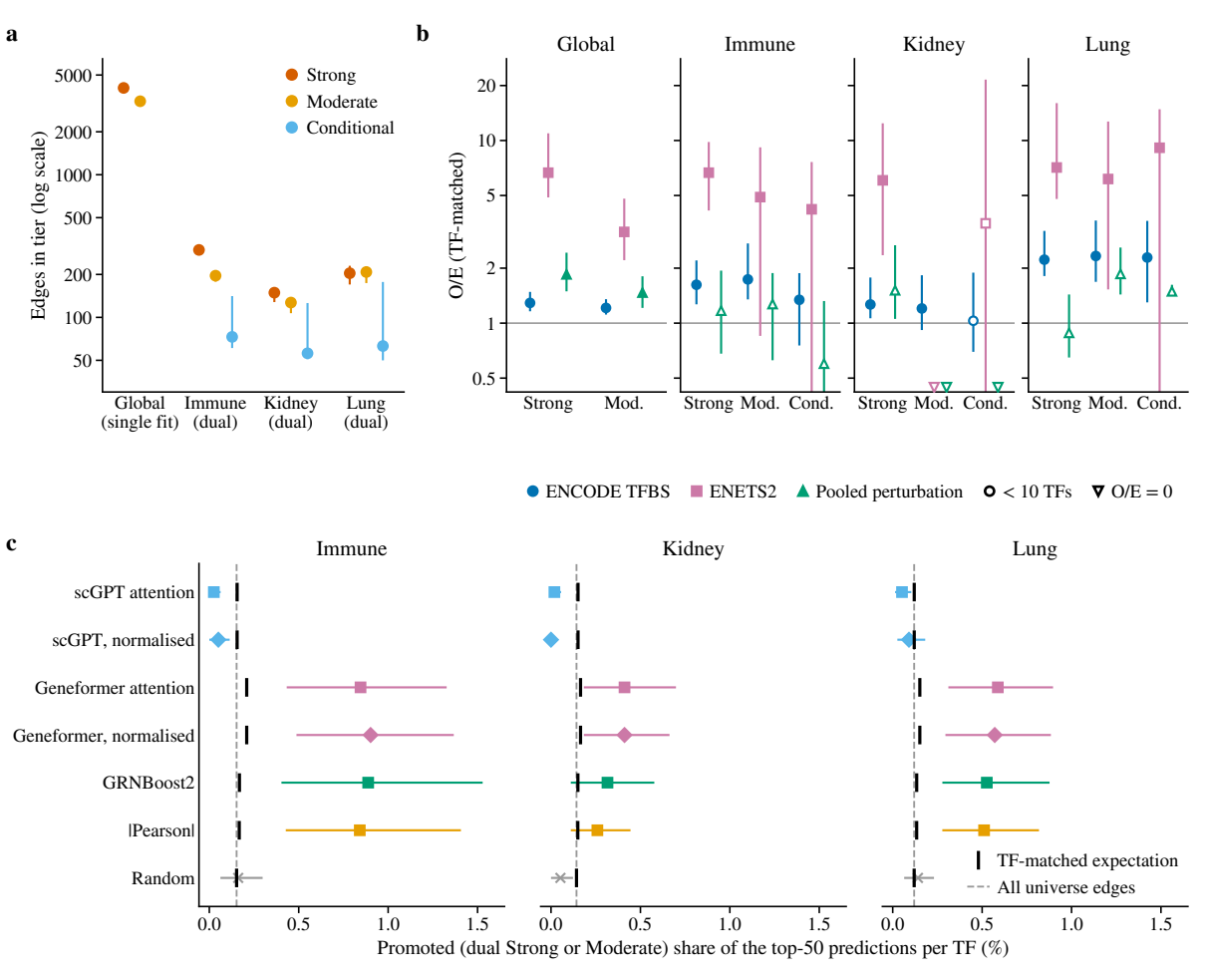
a
Edges in tier (log scale)
5000
2000
1000
500
200
100
50
Strong
Moderate
Conditional
Global (single fit)
Immune (dual)
Kidney (dual)
Lung (dual)
b
Global
Immune
Kidney
Lung
O/E (TF-matched)
20
10
5
2
1
0.5
Strong
Mod.
Cond.
ENCODE TFBS
ENETS2
Pooled perturbation
< 10 TFs
O/E = 0
c
Immune
Kidney
Lung
scGPT attention
scGPT, normalised
Geneformer attention
Geneformer, normalised
GRNBoost2
|Pearson|
Random
TF-matched expectation
All universe edges
0.0
0.5
1.0
1.5
Promoted (dual Strong or Moderate) share of the top-50 predictions per TF (%)